C. Local backend startup — commands to run

Open separate terminals.

1. Auth

cd C:\Users\HP\Desktop\smart-healthcare-microservices\backend\services\auth-service
npm run dev

Expected:

- MongoDB connected
- auth-service running on :4001

2. Patient

cd C:\Users\HP\Desktop\smart-healthcare-microservices\backend\services\patient-service
npm run dev

Expected:

- MongoDB connected
- patient-service running on :4002

3. Doctor

cd C:\Users\HP\Desktop\smart-healthcare-microservices\backend\services\doctor-service
npm run dev

Expected:

- MongoDB connected
- doctor-service running on :4003

4. Appointment

cd
C:\Users\HP\Desktop\smart-healthcare-microservices\backend\services\appointment-service
npm run dev

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

Expected:

- MongoDB connected
- appointment-service running on :4004

5. Payment

cd C:\Users\HP\Desktop\smart-healthcare-microservices\backend\services\payment-service
npm run dev

Expected:

- MongoDB connected
- payment-service running on :4007

6. Notification

cd
C:\Users\HP\Desktop\smart-healthcare-microservices\backend\services\notification-service
npm run dev

Expected:

- notification-service running on :4006

7. Telemedicine

cd
C:\Users\HP\Desktop\smart-healthcare-microservices\backend\services\telemedicine-service
npm run dev

Expected:

- telemedicine-service running on :4005

D. Quick health-check test in browser or Postman

Test these first.

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluster0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w=majority
JWT_SECRET=supersecret
```

GET

http://localhost:4001/health
http://localhost:4002/health
http://localhost:4003/health
http://localhost:4004/health
http://localhost:4005/health
http://localhost:4006/health
http://localhost:4007/health

All should return success JSON.

If any one fails, stop and fix that service before going further.

E. Postman — what to test now after Day 5

Because DB split changed things, do a **fresh run** in this exact order.

Use your environment variables and scripts as before.

FOLDER 1 — AUTH FLOW

1. Register Patient

Method

POST

URL

{{authUrl}}/auth/register

Body

{
 "name": "Patient Day5",

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

```
 "email": "patientday5@test.com",
 "password": "123456",
 "role": "PATIENT"
}
```

Tests

```
const res = pm.response.json();
pm.environment.set("patientUserId", res.userId);
```

---

2. Register Doctor

Method

POST

URL

{{authUrl}}/auth/register

Body

```
{
 "name": "Doctor Day5",
 "email": "doctorday5@test.com",
 "password": "123456",
 "role": "DOCTOR"
}
```

Tests

```
const res = pm.response.json();
pm.environment.set("doctorUserId", res.userId);
```

---

3. Register Admin

Method

POST

URL

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

{{authUrl}}/auth/register

Body

```
{
 "name": "Admin Day5",
 "email": "adminday5@test.com",
 "password": "123456",
 "role": "ADMIN"
}
```

Tests

```
const res = pm.response.json();
pm.environment.set("adminUserId", res.userId);
```

---

4. Login Admin

Method

POST

URL

{{authUrl}}/auth/login

Body

```
{
 "email": "adminday5@test.com",
 "password": "123456"
}
```

Tests

```
const res = pm.response.json();
pm.environment.set("adminToken", res.token);
pm.environment.set("adminUserId", res.userId);
```

---

5. Get Pending Doctors

Method

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

GET

URL

{{authUrl}}/auth/doctors/pending

Authorization

Bearer `{{adminToken}}`

Expected:

- your doctor should be listed

---

6. Verify Doctor

Method

PATCH

URL

{{authUrl}}/auth/doctors/{{doctorUserId}}/verify

Authorization

Bearer `{{adminToken}}`

Expected:

- success message
- doctorVerified true

---

7. Login Doctor

Method

POST

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

URL

{{authUrl}}/auth/login

Body

```
{
 "email": "doctorday5@test.com",
 "password": "123456"
}
```

Tests

```
const res = pm.response.json();
pm.environment.set("doctorToken", res.token);
pm.environment.set("doctorUserId", res.userId);
```

Expected:

- login success

---

8. Login Patient

Method

POST

URL

{{authUrl}}/auth/login

Body

```
{
 "email": "patientday5@test.com",
 "password": "123456"
}
```

Tests

```
const res = pm.response.json();
pm.environment.set("patientToken", res.token);
pm.environment.set("patientUserId", res.userId);
```

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluster0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w=majority
JWT_SECRET=supersecret
```

Expected:

- login success

FOLDER 2 — DOCTOR FLOW

9. Create/Get Doctor Profile

Method

GET

URL

{{doctorUrl}}/doctors/me

Authorization

Bearer `{{doctorToken}}`

Expected:

- doctor profile JSON
- auto-creates if missing

---

10. Update Doctor Profile

Method

PUT

URL

{{doctorUrl}}/doctors/me

Authorization

Bearer `{{doctorToken}}`

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluster0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w=majority
JWT_SECRET=supersecret
```

Body

```
{
 "specialty": "General Medicine",
 "bio": "Day 5 doctor profile"
}
```

Expected:

- updated profile returned

---

11. Update Availability

Method

PUT

URL

{{doctorUrl}}/doctors/me/availability

Authorization

Bearer `{{doctorToken}}`

Body

```
{
 "availability": [
   { "day": "MON", "from": "09:00", "to": "12:00" },
   { "day": "WED", "from": "14:00", "to": "17:00" }
 ]
}
```

Expected:

- availability saved

---

12. Get Doctors List

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

Method

GET

URL

{{doctorUrl}}/doctors

Expected:

- doctor profile appears in list

---

13. Get Single Doctor

Method

GET

URL

{{doctorUrl}}/doctors/{{doctorUserId}}

Expected:

- one doctor profile returned

FOLDER 3 — PATIENT FLOW

14. Get My Profile

Method

GET

URL

{{patientUrl}}/patients/me

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

Authorization

Bearer `{{patientToken}}`

Expected:

- patient profile JSON

---

15. Update My Profile

Method

PUT

URL

{{patientUrl}}/patients/me

Authorization

Bearer `{{patientToken}}`

Body

```
{
 "dateOfBirth": "2002-01-10",
 "gender": "Female",
 "phone": "0771234567",
 "address": "Colombo",
 "medicalHistory": "Asthma history",
 "allergies": ["Dust", "Seafood"],
 "chronicConditions": ["Asthma"]
}
```

Expected:

- updated profile saved

---

16. Upload Report

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

Method

POST

URL

{{patientUrl}}/patients/me/reports

Authorization

Bearer `{{patientToken}}`

Body

form-data:

- key: `report`
- type: File
- choose PDF/JPG/PNG

Tests

```
const res = pm.response.json();
if (res.file) pm.environment.set("reportFilename", res.file);
```

Expected:

- upload success
- filename saved

---

17. Get My Reports

Method

GET

URL

{{patientUrl}}/patients/me/reports

Authorization

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluster0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w=majority
JWT_SECRET=supersecret
```

Bearer `{{patientToken}}`

Expected:

- report list
- `downloadUrl` present

FOLDER 4 — APPOINTMENT FLOW

18. Create Appointment

Method

POST

URL

{{appointmentUrl}}/appointments

Authorization

Bearer `{{patientToken}}`

Body

```
{
 "doctorId": "{{doctorUserId}}",
 "datetime": "2026-04-01T10:00:00.000Z",
 "reason": "Fever and headache"
}
```

Tests

```
const res = pm.response.json();
pm.environment.set("appointmentId", res._id);
```

Expected:

- appointment created
- status `PENDING`

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluster0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w=majority
JWT_SECRET=supersecret
```

---

19. Get My Appointments

Method

GET

URL

{{appointmentUrl}}/appointments/me

Authorization

Bearer `{{patientToken}}`

Expected:

- appointment visible

---

20. Get Doctor Appointments

Method

GET

URL

{{appointmentUrl}}/appointments/doctor/me

Authorization

Bearer `{{doctorToken}}`

Expected:

- same appointment visible to doctor

---

21. Accept Appointment

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

Method

PUT

URL

{{appointmentUrl}}/appointments/{{appointmentId}}/status

Authorization

Bearer `{{doctorToken}}`

Body

```
{
 "status": "ACCEPTED"
}
```

Expected:

- appointment status becomes `ACCEPTED`
- telemedicine link created

---

22. Get Single Appointment

Method

GET

URL

{{appointmentUrl}}/appointments/{{appointmentId}}

Authorization

Bearer `{{patientToken}}`

Expected:

- `status: ACCEPTED`
- `telemedicineLink` present

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

FOLDER 5 — PAYMENT FLOW

23. Stripe Checkout Session

Method

POST

URL

{{paymentUrl}}/payments/checkout-session

Authorization

Bearer `{{patientToken}}`

Body

```
{
 "appointmentId": "{{appointmentId}}",
 "amount": 1000
}
```

Expected:

- `checkoutUrl`
- `paymentId`
- Stripe session id

If this fails, Avindya's Stripe config/env is the issue.

---

24. Fallback Demo Payment Create

Only if Stripe is not ready yet.

Method

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

POST

URL

{{paymentUrl}}/payments/for-appointment

Authorization

Bearer `{{patientToken}}`

Body

```
{
 "appointmentId": "{{appointmentId}}",
 "amount": 1000
}
```

Tests

```
const res = pm.response.json();
pm.environment.set("paymentId", res.paymentId);
```

Expected:

- payment created

---

25. Fallback Mark Paid

Only if using fallback demo path.

Method

POST

URL

{{paymentUrl}}/payments/mark-paid

Authorization

Bearer `{{patientToken}}`

Body

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

```
{
 "paymentId": "{{paymentId}}"
}
```

Expected:

- payment becomes `PAID`
- appointment becomes `CONFIRMED`

Important:
This works only if appointment is still `ACCEPTED`.

---

26. Get My Payments

Method

GET

URL

{{paymentUrl}}/payments/me

Authorization

Bearer `{{patientToken}}`

Expected:

- payment list visible

---

27. Payment Summary

Method

GET

URL

{{paymentUrl}}/payments/summary

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

Authorization

Bearer `{{adminToken}}`

Expected:

- summary JSON returns

FOLDER 6 — PRESCRIPTION FLOW

28. Create Prescription

Method

POST

URL

{{doctorUrl}}/prescriptions

Authorization

Bearer `{{doctorToken}}`

Body

```
{
 "patientId": "{{patientUserId}}",
 "appointmentId": "{{appointmentId}}",
 "meds": "Paracetamol 500mg twice daily",
 "notes": "Drink plenty of water"
}
```

Expected:

- prescription created

---

29. Get My Prescriptions as Patient

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

Method

GET

URL

{{patientUrl}}/patients/me/prescriptions

Authorization

Bearer `{{patientToken}}`

Expected:

- prescription visible

---

30. Get Doctor Issued Prescriptions

Method

GET

URL

{{doctorUrl}}/prescriptions/doctor/me

Authorization

Bearer `{{doctorToken}}`

Expected:

- issued prescription visible

FOLDER 7 — COMPLETE FLOW

31. Complete Appointment

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

Method

PATCH

URL

{{appointmentUrl}}/appointments/{{appointmentId}}/complete

Authorization

Bearer `{{doctorToken}}`

Expected:

- appointment status becomes `COMPLETED`

F. Kubernetes-specific checks after Day 5 YAML updates

Now the K8s part.

1. Apply the updated files

kubectl apply -f k8s/secret.yaml
kubectl apply -f k8s/configmap.yaml
kubectl apply -f k8s/auth.yaml
kubectl apply -f k8s/patient.yaml
kubectl apply -f k8s/doctor.yaml
kubectl apply -f k8s/appointment.yaml
kubectl apply -f k8s/payment.yaml
kubectl apply -f k8s/notification.yaml
kubectl apply -f k8s/telemedicine.yaml
kubectl apply -f k8s/ingress.yaml

2. Restart deployments

kubectl rollout restart deployment auth
kubectl rollout restart deployment patient
kubectl rollout restart deployment doctor

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

kubectl rollout restart deployment appointment
kubectl rollout restart deployment payment
kubectl rollout restart deployment notification
kubectl rollout restart deployment telemedicine

3. Check pods

kubectl get pods

What you want:

- all pods eventually `Running`
- no `CrashLoopBackOff`
- no `Error`

4. Check services

kubectl get svc

5. Check ingress

kubectl get ingress

6. If any pod fails, check logs

kubectl logs <pod-name>

For previous crash:

kubectl logs <pod-name> --previous

G. What exactly you should confirm after Day 5

DB split confirmation

You should confirm:

- auth-service writes to `smart-auth`

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluster0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w=majority
JWT_SECRET=supersecret
```

- patient-service writes to `smart-patient`
- doctor-service writes to `smart-doctor`
- appointment-service writes to `smart-appointment`
- payment-service writes to `smart-payment`

How to confirm

After doing the Postman flow, open Atlas and check:

- user collection appears in `smart-auth`
- patient profiles appear in `smart-patient`
- doctor profiles/prescriptions appear in `smart-doctor`
- appointments appear in `smart-appointment`
- payments appear in `smart-payment`

That is the whole point of Day 5 DB separation.

Yes — this is fixable, and the simplest way is to **throw away the broken local conflict state**, then pull fresh.

Vitharka's problem is now just:

- her local `doctor-service/package-lock.json` and `package.json` got messy
- Git is blocking `git pull` because of local changes
- GitHub still shows the PR merge conflict

Do this exactly

Part 1 — save anything important first

If she changed any real code in `doctor.routes.js` or other files today that is **not pushed**, tell her to copy that code somewhere safe first.

If not, continue.

Part 2 — go to the project root

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

She must be in:

C:\Users\vitha\smart-healthcare-microservices

Part 3 — hard reset the local branch state

Run these commands exactly:

git checkout feature/doctor-telemedicine
git fetch origin
git reset --hard origin/feature/doctor-telemedicine
git clean -fd

What this does

- makes local feature branch exactly same as GitHub branch
- removes broken local uncommitted changes
- removes the dirty state that is blocking pull

Part 4 — now update `dev`

Run:

git checkout dev
git fetch origin
git reset --hard origin/dev

Now local `dev` becomes exactly same as remote `dev`.

Part 5 — go back to her feature branch

Run:

git checkout feature/doctor-telemedicine
git merge dev

At this point, **if Git again says package-lock conflict**, do this:

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

del backend\services\doctor-service\package-lock.json
cd backend\services\doctor-service
npm install
cd ..\..\..
git add backend/services/doctor-service/package-lock.json
git commit -m "fix(doctor): regenerate package-lock after merge conflict"
git push origin feature/doctor-telemedicine

Even simpler version

Tell her to copy-paste this whole block from project root:

git checkout feature/doctor-telemedicine
git fetch origin
git reset --hard origin/feature/doctor-telemedicine
git clean -fd
git checkout dev
git fetch origin
git reset --hard origin/dev
git checkout feature/doctor-telemedicine
git merge dev

If merge stops on `package-lock.json`, then run:

del backend\services\doctor-service\package-lock.json
cd backend\services\doctor-service
npm install
cd ..\..\..
git add backend/services/doctor-service/package-lock.json
git commit -m "fix(doctor): regenerate package-lock after merge conflict"
git push origin feature/doctor-telemedicine

Why this happened

Because she:

- deleted and regenerated `package-lock.json`
- committed it on feature branch
- then switched branches with local modified files still hanging around

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluster0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w=majority
JWT_SECRET=supersecret
```

- then `dev` pull got blocked
- meanwhile GitHub PR still saw the merge conflict

So the fix is:

reset local branches cleanly, then merge again cleanly

Important note

She used:

rm package-lock.json

That failed because Windows CMD does not use `rm`.

In **CMD**, use:

del package-lock.json

In **PowerShell**, use:

Remove-Item package-lock.json

What success looks like

At the end:

- `git status` should say clean
- `git merge dev` should complete
- if conflict happened, regenerated `package-lock.json` should fix it
- `git push origin feature/doctor-telemedicine` should update branch
- GitHub PR conflict banner should disappear

If she wants the safest route of all

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```

If this still acts weird, do this nuclear-safe option:

from project root

```
git checkout feature/doctor-telemedicine
git fetch origin
git reset --hard origin/feature/doctor-telemedicine
del backend\services\doctor-service\package-lock.json
cd backend\services\doctor-service
npm install
cd ..\..\..
git add backend/services/doctor-service/package-lock.json
git commit -m "fix(doctor): regenerate package-lock cleanly"
git push origin feature/doctor-telemedicine
git fetch origin
git merge origin/dev
git push origin feature/doctor-telemedicine
```

Send me the exact output of `git merge dev` after she runs the first block

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluster0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w=majority
JWT_SECRET=supersecret
```

```
PORT=4002
MONGO_URI=mongodb+srv://smartuser:StrongPass123!@cluste
r0.lpgj8zl.mongodb.net/smart-patient?retryWrites=true&w
=majority
JWT_SECRET=supersecret
```